

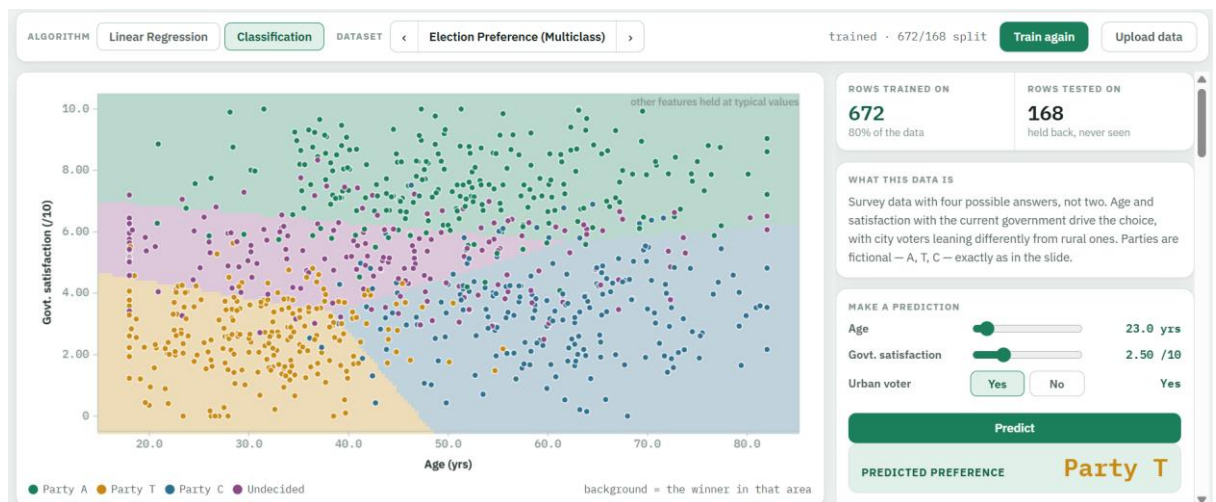
<https://testleaf-ai-playground.sudarshan210688.chatgpt.site/Home> Assignment – Week1Day1

1) Classification Algorithm

- a) Age <30 and Govt satisfaction less, urban
- b) Age >50 and Govt satisfaction less, non urban

Validate the percentage

- a) Age <30 and Govt satisfaction less, urban



Party A

0.0%

Party T

95.4%

Party C

1.9%

Undecided

2.8%

Step 1 – each class gets its own score = base
+ (weight × this voter's value), added up

Party A

-12.4 pts ← base points

+ 0.028 × 34.0 (Age) = +0.958

+ 2.11 × 2.50 (Govt. satisfaction) = +5.28

+ 0.150 × Yes (Urban voter) = +0.150

= -6.04 pts

Party T

10.3 pts $\leftarrow$ base points

– **0.123** $\times$ **34.0** (Age) = **-4.17**

– **1.43** $\times$ **2.50** (Govt. satisfaction) = **-3.57**

+ **2.31** $\times$ **Yes** (Urban voter) = **+2.31**

= 4.88 pts

Party C

-0.378 pts $\leftarrow$ base points

+ **0.105** $\times$ **34.0** (Age) = **+3.55**

– **0.840** $\times$ **2.50** (Govt. satisfaction) =
-2.10

– **0.127** $\times$ **Yes** (Urban voter) = **-0.127**

= 0.949 pts

Undecided

0.139 pts $\leftarrow$ base points

– **0.020** $\times$ **34.0** (Age) = **-0.670**

Undecided

$$\begin{aligned} & 0.139 \text{ pts} \leftarrow \text{base points} \\ & - 0.020 \times 34.0 \text{ (Age)} = -0.670 \\ & + 0.421 \times 2.50 \text{ (Govt. satisfaction)} = \\ & +1.05 \\ & + 0.823 \times \text{Yes (Urban voter)} = +0.823 \\ & \hline & = 1.34 \text{ pts} \end{aligned}$$

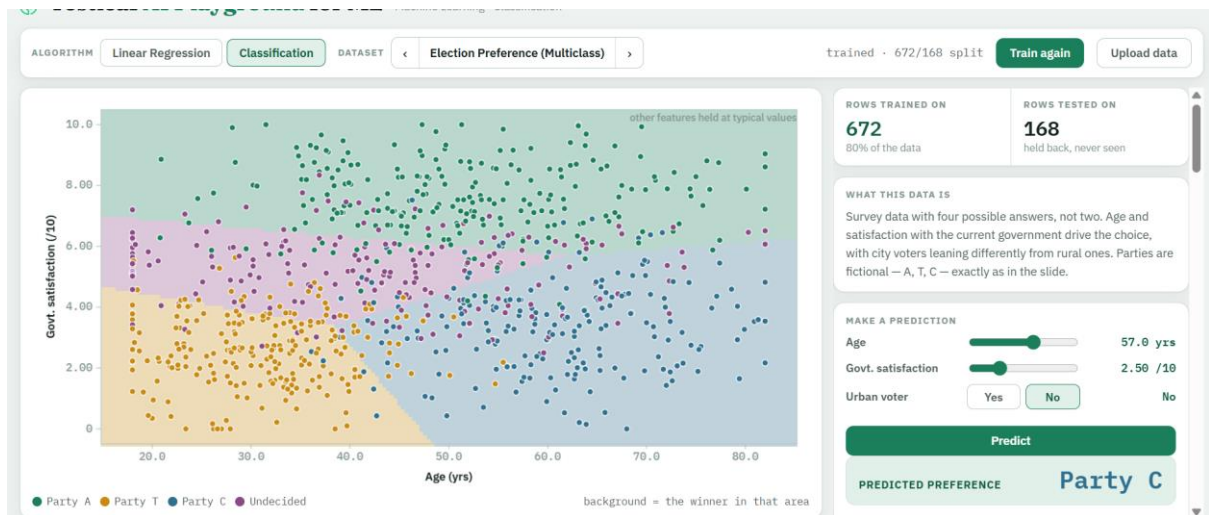
highest score wins → Party T

Step 2a – convert each score to e^{score}
(always positive)

$$\begin{aligned} \text{Party A} & \quad e^{-6.04} = 0.002 \\ \text{Party T} & \quad e^{4.88} = 132 \\ \text{Party C} & \quad e^{0.949} = 2.58 \\ \text{Undecided} & \quad e^{1.34} = 3.83 \end{aligned}$$

$$\text{Total (add all 4 up)} = 139$$

b) Age > 50 and Govt satisfaction less, non urban



Predict

PREDICTED PREFERENCE

Party C

Party A

0.0%

Party T

2.3%

Party C

94.6%

Undecided

3.1%

Step 1 – each class gets its own score = base
+ (weight × this voter's value), added up

Party A

-12.4 pts ← base points

+ 0.028 × 57.0 (Age) = +1.61

+ 2.11 × 2.50 (Govt. satisfaction) = +5.28

$$= -5.54 \text{ pts}$$

Party T

$$10.3 \text{ pts} \leftarrow \text{base points}$$

$$- 0.123 \times 57.0 \text{ (Age)} = -6.99$$

$$- 1.43 \times 2.50 \text{ (Govt. satisfaction)} = -3.57$$

$$+ 2.31 \times \text{No (Urban voter)} = +0$$

$$= -0.243 \text{ pts}$$

Party C

$$-0.378 \text{ pts} \leftarrow \text{base points}$$

$$+ 0.105 \times 57.0 \text{ (Age)} = +5.96$$

$$- 0.840 \times 2.50 \text{ (Govt. satisfaction)} = -2.10$$

$$- 0.127 \times \text{No (Urban voter)} = +0$$

$$= 3.48 \text{ pts}$$

Undecided

$$\begin{aligned}
& 0.139 \text{ pts} \leftarrow \text{base points} \\
& - 0.020 \times 57.0 \text{ (Age)} = -1.12 \\
& + 0.421 \times 2.50 \text{ (Govt. satisfaction)} = \\
& +1.05 \\
& + 0.823 \times \text{No (Urban voter)} = +0 \\
& \hline
& = 0.068 \text{ pts}
\end{aligned}$$

highest score wins → Party C

Step 2a – convert each score to e^{score}
(always positive)

Party A	$e^{-5.54} = 0.004$
Party T	$e^{-0.243} = 0.785$
Party C	$e^{3.48} = 32.5$
Undecided	$e^{0.068} = 1.07$

$$\text{Total (add all 4 up)} = 34.4$$

SAMPLE CALCULATION — ONE REAL ROW

45.7 yrs · 6.43 /10 · not urban voter

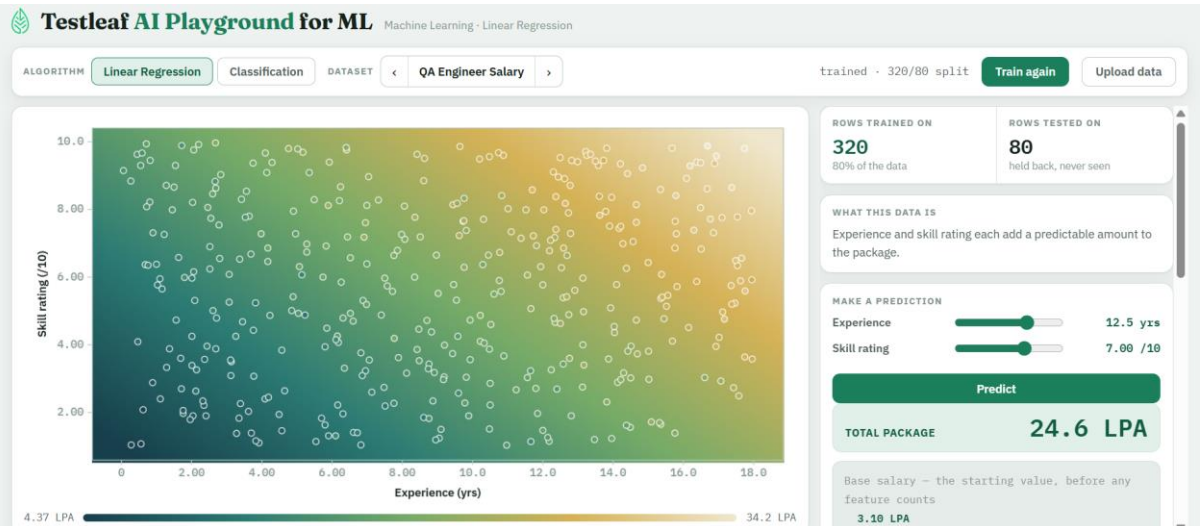
Party A	2.44
Party T	-4.47
Party C	-1.00
Undecided	1.94

highest wins → Party A

actual Party A

With more than two possible answers, the model keeps **one score per class** — four separate sums from the same inputs. Whichever score comes out highest is the prediction. That is all “multiclass” means.

- 2) Linear Regression
 - a) Qa salary
 - b) Food Delivery Time
- ** QA salary



Base salary – the starting value, before any feature counts

3.10 LPA

Experience

0.966 LPA per year × 12.5 yrs

= +12.1 LPA

Skill rating

1.34 LPA per point × 7.00 /10

= +9.39 LPA

TOTAL 24.6 LPA

THE TRAINED MODEL

Salary_LPA in LPA =

3.10 LPA ← base salary (minimum)
+ **0.966** LPA per year × Experience
+ **1.34** LPA per point × Skill rating

SAMPLE CALCULATION — ONE REAL ROW FROM THE DATA

9.46 yrs · 5.78 /10

3.10 (base)
+ **0.966** × **9.46** = **+9.14**
+ **1.34** × **5.78** = **+7.75**

model says 20.0 LPA
actual value 19.4 LPA

Read it as a rate card. The first number is the **base salary** — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.

c) Food Delivery Time



ALGORITHM

Linear Regression

Classification

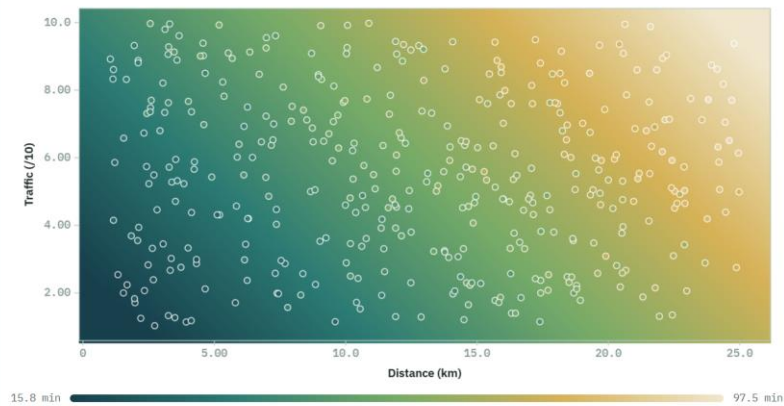
DATASET

Food Delivery Time

trained · 320/80 split

Train again

Upload data



ROWS TRAINED ON

320

80% of the data

ROWS TESTED ON

80

held back, never seen

WHAT THIS DATA IS

Distance and traffic both push the delivery time up at a steady rate.

MAKE A PREDICTION

Distance

20.0 km

Traffic

4.90 /10

Predict

TOTAL TIME

71.1 min

Base time – the starting value, before any feature counts

7.56 min

Base time – the starting value, before any feature counts

7.56 min

Distance

2.40 min per km × 20.0 km

= +48.0 min

Traffic

3.16 min per point × 4.90 /10

= +15.5 min

TOTAL 71.1 min

THE TRAINED MODEL

Delivery_Minutes in min =

THE TRAINED MODEL

Delivery_Minutes in min =

7.56 min ← base time (minimum)
+ **2.40** min per km × Distance
+ **3.16** min per point × Traffic

SAMPLE CALCULATION — ONE REAL ROW FROM THE DATA

8.96 km · 8.46 /10

7.56 (base)
+ **2.40** × **8.96** = **+21.5**
+ **3.16** × **8.46** = **+26.7**

model says 55.8 min
actual value 55.4 min

Read it as a rate card. The first number is the **base time** — the minimum you start from before any feature is counted. Every line after it says what one more unit is worth. Press Predict to see it added up.